ON-BOARD DIAGNOSTIC [FORWARD SENSING CAMERA (FSC)]

id092200021852

Outline

- The on-board diagnostic function consists of the following functions:
  - — A malfunction detection function, which detects overall malfunctions in the forward sensing camera (FSC)-related parts.
  - — A memory function, which stores detected DTCs.
  - — A display function, which indicates malfunction locations and status via DTC output.
  - — A PID/data monitoring function, which reads out specific input/output signals and verifies the input/output condition.
- Using the Mazda Modular Diagnostic System (M-MDS), DTCs can be read out and deleted, and the PID/data monitoring function can be activated.

Malfunction detection function

- Detects malfunctions in input/output signals.
- If a malfunction occurs, the forward sensing camera (FSC) records the malfunction as a DTC. A recorded DTC can be read by the Mazda Modular Diagnostic System (M-MDS).

DTC table

X: Applicable
Not applicable

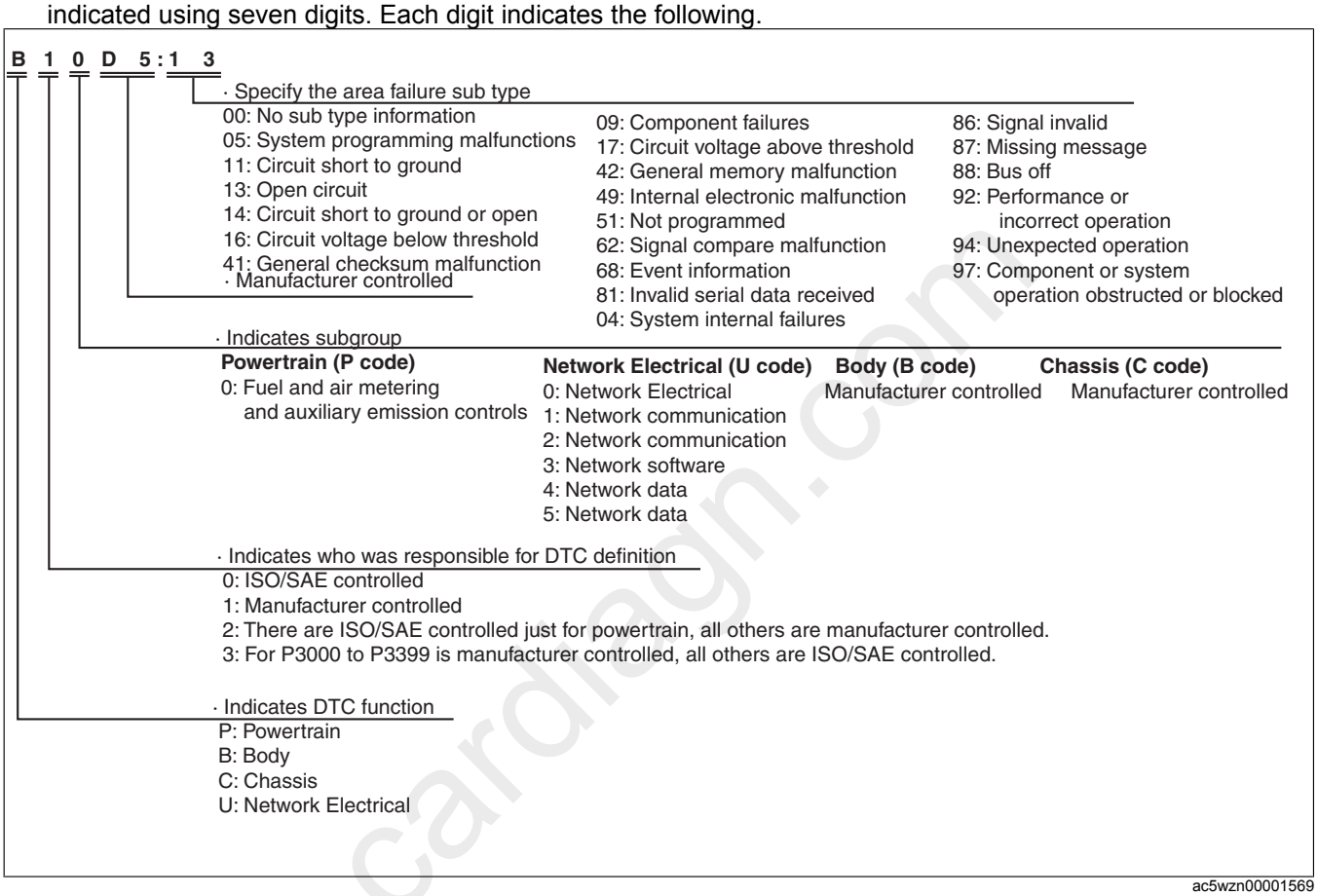
DTC No.	LDWS warning light	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
B115E:54	Illuminated	Forward sensing camera (FSC) aiming not completed	×	—	C, D	×
B13A6:11	Illuminated	LDWS switch circuit malfunction	×	—	C, D	×
C1001:92	Illuminated	Poor functioning of forward sensing camera (FSC)	×	—	C, D	×
C1001:97	Illuminated	Forward sensing camera (FSC) poor visibility	×	—	C, D	×
U0001:88	Illuminated	Module communication error (HS-CAN)	×	—	C, D	×
U0100:00	Illuminated	Communication error with PCM	×	—	C, D	×
U0121:00	Illuminated	Communication error with DSC HU/CM	×	—	C, D	×
U0131:00	Illuminated	Communication error with EPS control module	×	—	C, D	×
U0140:00	Illuminated	Communication error with BCM	×	—	C, D	×
U0155:00	Illuminated	Communication error with instrument cluster	×	—	C, D	×
U0214:00	Illuminated	Communication error with start stop unit	×	—	C, D	×
U0401:68	Illuminated	Error signal received from PCM	×	—	C, D	×
U0415:68	Illuminated	Error signal received from DSC HU/CM	×	—	C, D	×
U0420:68	Illuminated	Error signal received from EPS control module	×	—	C, D	×
U0423:68	Illuminated	Error signal received from instrument cluster	×	—	C, D	×
U0515:68	Illuminated	Error signal received from start stop unit	×	—	C, D	×
U2005:86	Illuminated	Error signal received from PCM	×	—	C, D	×
U2100:00	Illuminated	Forward sensing camera (FSC) configuration error	×	—	C, D	×
U2101:00	Illuminated	Forward sensing camera (FSC) configuration error	×	—	C, D	×
U3000:04	Illuminated	Forward sensing camera (FSC) internal malfunction	×	—	C, D	×
U3000:09	Illuminated	Forward sensing camera (FSC) internal malfunction	×	—	C, D	×
U3000:42	Illuminated	Forward sensing camera (FSC) internal malfunction	×	—	C, D	×
U3000:49	Illuminated	Forward sensing camera (FSC) internal malfunction	×	—	C, D	×
U3003:16	Illuminated	Forward sensing camera (FSC) low power supply voltage input	×	—	C, D	×

DTC No.	LDWS warning light	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
U3003:17	Illuminated	Forward sensing camera (FSC) high power supply voltage input	×	—	C, D	×

*1 : C: CMDTC self test, D: ODDTC self test

DTC 7-digit code definition

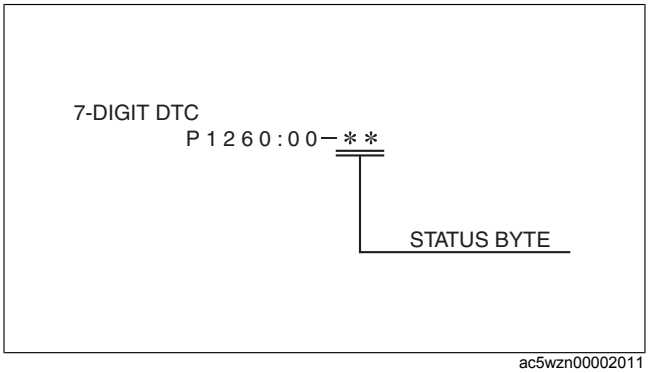
- When related systems or components have failed, the CM stores the DTC of the malfunctioning part in the CM memory, and allows for the retrieval of the store data using scanning tool when necessary. The DTCs are indicated using seven digits. Each digit indicates the following.



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Status byte for DTC

- The two digits (two digits after hyphen (-)) after the 7-digit DTC.
- A code which indicates the pending code, current/ past malfunction status, or warning illumination status.
- Can be read by performing a CMDTC self-test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the Mazda Modular Diagnostic System (M-MDS) when reading the DTC.



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Detection condition for the applicable DTC

DTC No.	Description	Detection condition
B115E:54	Forward sensing camera (FSC) aiming not completed	• Forward sensing camera (FSC) aiming not completed is detected.
B13A6:11	LDWS switch circuit malfunction	• Forward sensing camera (FSC) detects that LDWS switch is held pressed for **10 s or more**.

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DTC No.	Description	Detection condition
C1001:92	Poor functioning of forward sensing camera (FSC)	• With the LDWS switch on or the high beam control (HBC) system operating, the forward sensing camera (FSC) detects that its field of view is poor for a certain period of time.
C1001:97	Forward sensing camera (FSC) poor visibility	• With the LDWS switch on or the high beam control (HBC) system operating, the forward sensing camera (FSC) detects that its field of view is obstructed for a certain period of time.
U0001:88	Module communication error (HS-CAN)	• Forward sensing camera (FSC) cannot receive the CAN bus communication signal (HS-CAN) for **1 s or more**. • Forward sensing camera (FSC) detects the CAN bus communication line (HS-CAN) malfunction **10 times** continuously.
U0100:00	Communication error with PCM	• Forward sensing camera (FSC) cannot receive the CAN signal from the PCM for **1 s or more**.
U0121:00	Communication error with DSC HU/CM	• Forward sensing camera (FSC) cannot receive the CAN signal from the DSC HU/CM for **1 s or more**.
U0131:00	Communication error with EPS control module	• Forward sensing camera (FSC) cannot receive the CAN signal from the EPS control module for **1 s or more**.
U0140:00	Communication error with BCM	• Forward sensing camera (FSC) cannot receive the CAN signal from the BCM for **5 s or more**.
U0155:00	Communication error with instrument cluster	• Forward sensing camera (FSC) cannot receive the CAN signal from the instrument cluster for **5 s or more**.
U0214:00	Communication error with start stop unit	• Forward sensing camera (FSC) cannot receive the CAN signal from the start stop unit for **1 s or more**.
U0401:68	Error signal received from PCM	• Forward sensing camera (FSC) receives the error signal from the PCM for **5 s or more**.
U0415:68	Error signal received from DSC HU/CM	• Forward sensing camera (FSC) receives the error signal from the DSC HU/CM for **5 s or more**.
U0420:68	Error signal received from EPS control module	• Forward sensing camera (FSC) receives the error signal from the EPS control module for **5 s or more**.
U0423:68	Error signal received from instrument cluster	• Forward sensing camera (FSC) receives the error signal from the instrument cluster for **5 s or more**.
U0515:68	Error signal received from start stop unit	• Forward sensing camera (FSC) receives the error signal from the start stop unit for **5 s or more**.
U2005:86	Error signal received from PCM	• Forward sensing camera (FSC) receives the vehicle speed signal error from the PCM for **5 s or more**.
U2100:00	Forward sensing camera (FSC) configuration error	• Forward sensing camera (FSC) configuration not completed is detected.
U2101:00	Forward sensing camera (FSC) configuration error	• Detects that forward sensing camera (FSC) has unsupported configuration data.
U3000:04	Forward sensing camera (FSC) internal malfunction	• Forward sensing camera (FSC) detects an error in the image-related circuit in the forward sensing camera (FSC) with the ignition switched ON (engine off or on).
U3000:09	Forward sensing camera (FSC) internal malfunction	• Forward sensing camera (FSC) detects an internal temperature sensor malfunction in the forward sensing camera (FSC) with the ignition switched ON (engine off or on).
U3000:42	Forward sensing camera (FSC) internal malfunction	• Forward sensing camera (FSC) detects a NVRAM error in the forward sensing camera (FSC) with the ignition switched ON (engine off or on).
U3000:49	Forward sensing camera (FSC) internal malfunction	• Forward sensing camera (FSC) detects a malfunction in the forward sensing camera (FSC) with the ignition switched ON (engine off or on).
U3003:16	Forward sensing camera (FSC) low power supply voltage input	• Forward sensing camera (FSC) power supply circuit voltage of **8.0 ± 0.8 V or less** is detected for **5 s or more** with the ignition switched ON (engine off or on).

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DTC No.	Description	Detection condition
U3003:17	Forward sensing camera (FSC) high power supply voltage input	• Forward sensing camera (FSC) power supply circuit voltage of **16.0 ± 0.8 V or more** is detected for **5 s or more** with the ignition switched ON (engine off or on).

Snap shot data

Note
- The forward sensing camera (FSC) stores the following two types of snapshot data (vehicle information) when a DTC is detected and displays them in the M-MDS.
  - — Vehicle information detected by forward sensing camera (FSC)
  - — Vehicle information detected by instrument cluster and received by forward sensing camera (FSC) via CAN communication

- The data for all DTCs currently detected is stored.

—: Not applicable

Snapshot data item	Unit		Data contents	Data read/use method	Corresponding data monitor items
ECU_IN_TMP	°C	°F	ECU Internal Temperature	—	—
IG-ON_TIMER	hh:mm:ss*1		Elapsed time since ignition was switched ON (engine off or on) **Note** • The instrument cluster records the elapsed time since the ignition was switched ON (engine off or on).	• The forward sensing camera (FSC) constantly receives the elapsed time since the ignition was switched ON (engine off or on) sent via CAN communication from the instrument cluster. • If a DTC is detected, the forward sensing camera (FSC) records the elapsed time since the ignition was switched ON (engine off or on) when the DTC was detected, and it is displayed in the M-MDS.	—
OPE_MODE	Normal/ CALIBRATION/ EMERGENCY		Operation Mode	—	—
VSPD	KPH	MPH	Vehicle speed	—	—

*1 : The seconds may be indicated after the decimal point.

Data Monitor Function
- With the PID/data monitor function, the input/output signal monitor items set in the forward sensing camera (FSC) can be selected and read out in real-time.
- The items which can be selected are as follows.

PID/data monitor table

PID	Unit/ Operation	Data contents	Module control terminal
A_ANG_PITCH	° (deg)	Displays vanishing point correction angle (pitch) corrected by forward sensing camera (FSC) when performing forward sensing camera (FSC) aiming using Mazda Modular Diagnostic System (M-MDS).	C, E (CAN)
A_ANG_ROLL	° (deg)	Displays vanishing point correction angle (roll) corrected by forward sensing camera (FSC) when performing forward sensing camera (FSC) aiming using Mazda Modular Diagnostic System (M-MDS).	C, E (CAN)
A_ANG_YAW	° (deg)	Displays vanishing point correction angle (yaw) corrected by forward sensing camera (FSC) when performing forward sensing camera (FSC) aiming using Mazda Modular Diagnostic System (M-MDS).	C, E (CAN)

PID	Unit/ Operation	Data contents	Module control terminal
BRAKE_SW	Off/On	• Off: Brake switch (No. 2 signal) is off. (Brake pedal is not depressed.) • On: Brake switch (No. 2 signal) is on. (Brake pedal is depressed.)	C, E (CAN)
DSP_LANE_ST	Off/ Dash_Lane/ Solid_Lane	• Off: Lane departure warning system (LDWS) is off. • Dash_Lane: When any one of the following conditions is met: — Actual vehicle speed is less than vehicle speed for operation of lane departure warning system (LDWS). — Dirt or fogging of camera is detected. — Vehicle lane lines are not recognized. • Solid_Lane: Vehicle lane lines are recognized at a vehicle speed of **40 km/h {25 mph} or more**.	C, E (CAN)
DSP_WRN_ST	Off/ LDWS_Left/ LDWS_Right/ LDWS_Fail/ HBC_Fail/ Camera_BLK/FOW Warning/ Fow Fail/ FSC_Fail/ Camera_DRDT	• Off: When any one of the following conditions is met: — LDWS switch is off. — Lane departure warning system (LDWS) does not determine warning. — Forward sensing camera (FSC) does not detect malfunction. • LDWS_Left: Lane departure on vehicle left side is detected. • LDWS_Right: Lane departure on vehicle right side is detected. • LDWS_Fail: Malfunction related to lane departure warning system (LDWS) is detected. • HBC_Fail: Malfunction related to HBC system is detected. • Camera_BLK: Dirt on camera is detected. • FOW_Warning: — • FOW_Fail: — • FSC_Fail: Forward sensing camera (FSC) malfunction is detected. • Camera_DRDT: Camera fogging is detected.	C, E (CAN)
HBC_H/L_CS	Hi_Request/ Low_Request/Invalid	• Hi_Request: HI beam switch signal received • Low_Request: LO beam switch signal received • Invalid: Error in HI/LO beam switch request signal	C, E (CAN)
IL_ST_HBC	Off/ Green_On/ Green_Flash/ Amber_On/ Amber_Flash	• Off: HBC system indicator light (green) and HBC system warning indicator light (amber) turned off • Green_On: HBC system indicator light (green) is illuminated. • Green_Flash: HBC system indicator light (green) is flashing. • Amber_On: HBC system warning light (amber) is illuminated. • Amber_Flash: HBC system warning light (amber) is flashing.	C, E (CAN)
IL_ST_LDWS	Off/ Green_On/ Green_Flash/ Amber_On/ Amber_Flash	• Off: LDWS indicator light (green) and LDWS warning light (amber) turned off • Green_On: LDWS indicator light (green) illuminated • Green_Flash: LDWS indicator light (green) is flashing. • Amber_On: LDWS warning light (amber) is illuminated. • Amber_Flash: LDWS warning light (amber) is flashing.	C, E (CAN)
LDWS_BUZZER	Off/On	• Off: LDWS warning alarm does not sound. • On: LDWS warning alarm sounds.	C, E (CAN)
LDWS_SW	Off/On	• Off: LDWS switch is off. • On: LDWS switch is on.	C, E (CAN)
RS_SOUND_CS	Off/Left/ Right	• Off: Rumble strip sound is not heard. • Left: Left side rumble strip sound heard. • Right: Right side rumble strip sound heard.	C, E (CAN)
VPWR_IG1	V	Displays forward sensing camera (FSC) power supply voltage	I (Power supply)
VSPD	KPH, MPH	Displays vehicle speed	C, E (CAN)